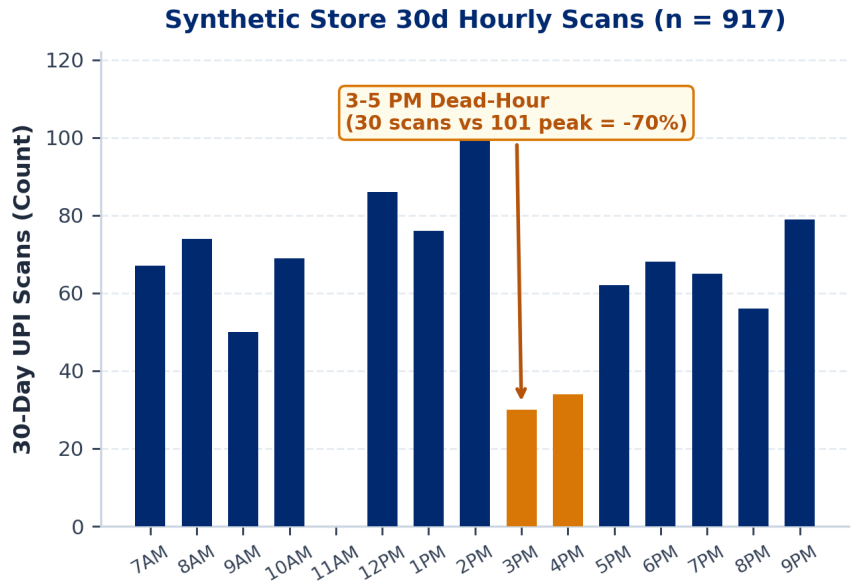
1.57 Cr Storefronts Accept QR Payments; Retention Stays Manual

* **Verified Paytm Scale:** 1.57 Cr merchant device subscribers and Rs 7.1L Cr quarterly GMV (Paytm Q1 FY27, Jul 2026).

* **Silent Repeat Churn:** Small shops lose an estimated 25% to 35% of repeat buyers yearly without alerts (Assumption).

* **No Time for Dashboards:** Counter operators work 12+ hr days; SMB BI apps see <8% weekly active usage (Assumption).

* **Dead-Hour & Prep Gaps:** Afternoon scans (3 to 5 PM) drop 70% vs peak in our synthetic store, hurting daily margins.



PaytmAI VyaparVoice (Track 1)

Proactive closing-time voice copilot for Paytm Soundbox & Business App.

Team SemiTechnical | Kevin Joseph

One Closing-Time Voice Action, Governed by Deterministic Rules

* **8:30 PM Voice Pitch:** Speaks one concrete action in Kannada, Hindi, or English (draft strings; native review pending).

* **Synthetic Store Demo:** Sri Krishna Darshini (240 payers, 1,613 UPI scans); detects 48 regulars silent 16+ days (median 19d).

* **5-Rule Policy Engine:** Excludes 5 unconsented (DPDP) and 3 on 14d cooldown; splits 40 eligible into 32 nudged / 8 control.

* **Money Guardrails:** Rs 25 cashback on Rs 150+ scan (16.7% <= 22% floor), Rs 300 cap; no LLM ever writes numbers or caps.

* **Ops & Care Actions:** Same 1-tap flow triggers simulated weekend dairy restock POs and queue-delay customer recovery.

Deterministic Core Built Today; Ecosystem Integrations Scoped

* **[BUILT IN MVP] FastAPI + SQLite Core:** Live SQL cohort queries, 5-rule policy checks, 80/20 holdout math, audit log, 21/21 QA tests.

* **[BUILT IN MVP] Companion PWA + Audio:** Paytm Soundbox simulator UI using browser speech synthesis + 1-tap button fallback (offline-safe).

* **[PENDING VERIFICATION] Sponsor Voice & Q&A:** Keyless adapters for Sarvam TTS/ASR (Saturday window) and free-tier Groq grounded English Q&A.

* **[PRODUCTION PLAN] Paytm Rail Hooks:** Paytm QR dynamic coupon ledger, Soundbox 4G push audio, WhatsApp Cloud POs, Postgres/Redis.

Simulated Holdout Lift & Profit Math (To Validate in Pilot)

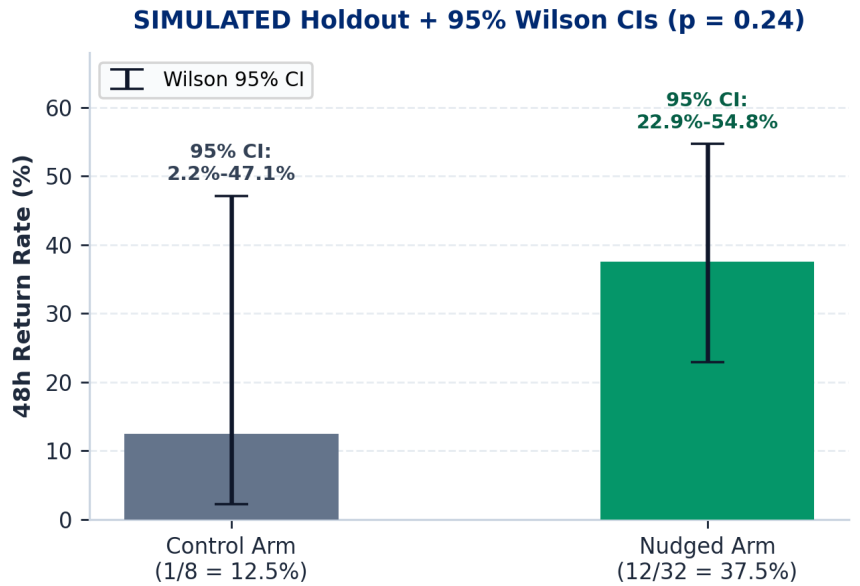
* **Simulated 48h Return:** 12/32 nudged (37.5%) vs 1/8 control (12.5%); Fisher exact p = 0.24 (not significant).

* **Pilot Sample Target:** 80% power at alpha = 0.05 needs ~46 payers/arm (~4 weekly store waves).

* **Sales Multiple (5.8x):** 8 net visits x Rs 217.50 assumed bill = Rs 1,740 incremental sales on Rs 300 spend.

* **Organic Subsidy:** 4 of 12 redeemers (12.5% baseline) would visit anyway, absorbing Rs 100 cashback.

* **Profit ROI (30% Margin):** Rs 522 gross profit (1.74x); +Rs 222 net pre-fee (+74% ROI), +Rs 91.50 post 5% fee.



EVIDENCE DISCLOSURE (SIMULATED)

Redemptions come from our deterministic 48h simulator. Break-even store gross margin is 17.2% pre-fee (24.7% after 5% Paytm promo fee).

Three Modelled Revenue Cases (5% Base) & 45-Day Pilot Plan

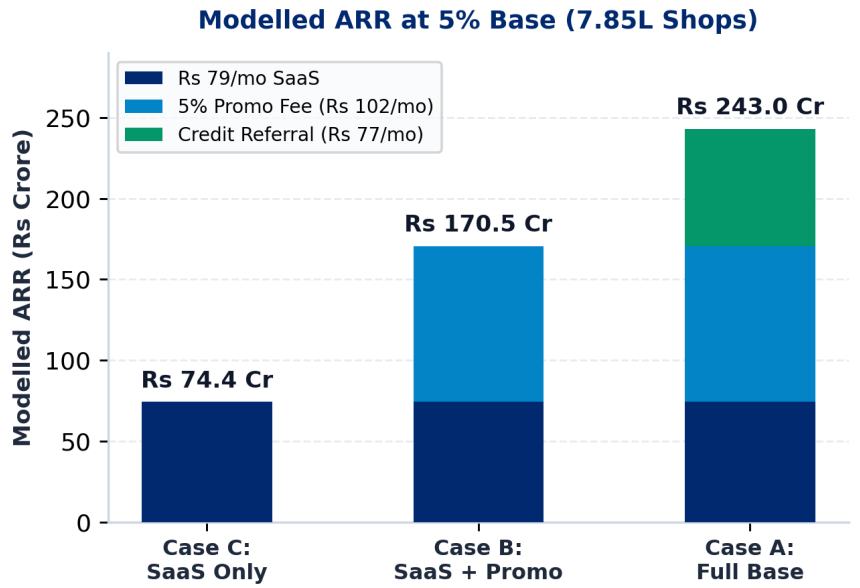
* **Case C (SaaS Only):** Rs 79/mo Soundbox add-on x 7.85L shops (5% base) = Rs 74.4 Cr ARR (Modelled).

* **Case B (Excl. Credit):** Adding 5% fee on redeemed bills (+Rs 102/mo) = Rs 170.5 Cr ARR (Modelled).

* **Case A (Full Base):** Adding partner loan referral (+Rs 77/mo; Paytm pen. ~7%, Q4 FY26) = Rs 243.0 Cr ARR.

* **Consistent Unit COGS:** Rs 48/mo cached (Rs 5 AI + Rs 43 msg; 81.4% margin) or Rs 71.80 uncached TTS (72.2%).

* **45-Day Bengaluru Pilot:** 50 kirana/F&B shops, >=50 payers/arm pooled holdout, native Kannada/Hindi sign-off.



TTS CACHE ASSUMPTION

Bulbul v3 lists Rs 30 / 10k chars (~Rs 27/mo uncached at 300 chars/day). Achieving Rs 3.20/mo TTS requires ~88% template caching (or v2 pricing).

APPENDIX A1 (CORRECTED ORIGINAL SLIDE 1) | TRACK 1: MERCHANT GROWTH AI

PaytmAI VyaparVoice

Proactive Voice Growth, Operations & Retention Copilot for Paytm Merchants

Designed for Paytm's 1.57 Cr merchant device base (Q1 FY27). Combines deterministic SQL cohort rules and budget guardrails with closing-time voice briefings (synthetic MVP built; live pilot pending).

01 : Voice + Button Flow

English, Kannada & Hindi draft scripts; 1-tap button fallback

02 : 5-Rule Policy Engine

DPDP opt-in, 14d cooldown, <=22% floor & Rs 300 cap

03 : Simulated Holdout

80/20 control split; 5.8x sales/promo (1.74x gross profit)

TEAM: SemiTechnical | Builder: Kevin Joseph (Bengaluru)

Appendix Status: Corrected from AUDIT.md (all synthetic, simulated, and pending items explicitly labelled).

Problem Details, Dated Market Sources & Synthetic Persona

1. Silent Repeat Churn (25% to 35% Yearly - ASSUMPTION)

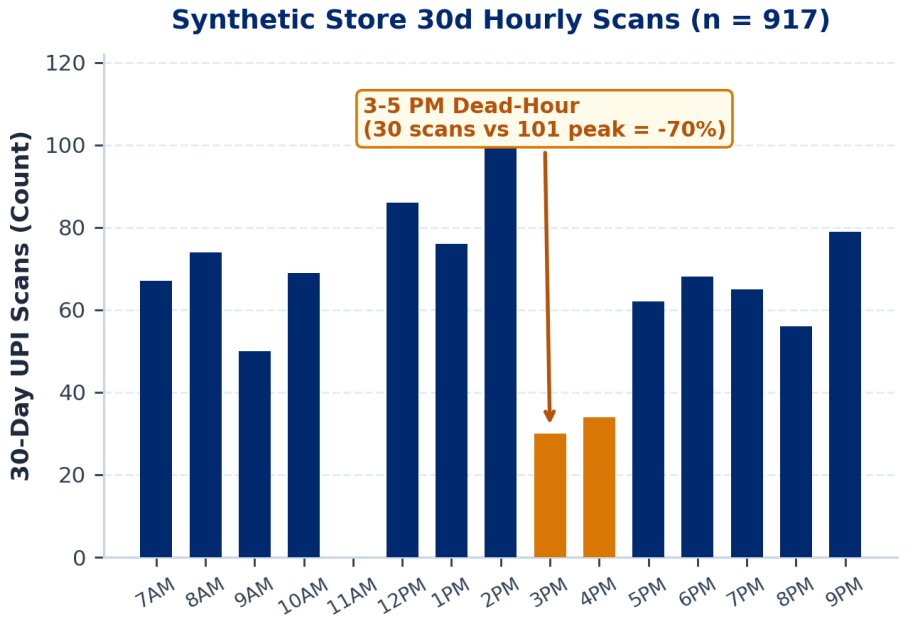
Merchants receive isolated payment alerts without customer-level retention tracking, losing repeat buyers before cashflow drops are noticed.

2. Dashboard Friction (<8% WAU - ASSUMPTION)

Counter-operated kirana, darshini, and pharmacy shops lack time or training to write prompts, export CSVs, or configure CRM rules.

3. Dead Hours & Rush Prep (SYNTHETIC STORE)

In our 60-day synthetic Indiranagar ledger, 3 to 5 PM scans fall 70% below peak while weekend morning rushes risk staple stockouts.



VERIFIED PAYTM DISCLOSURES vs SYNTHETIC PERSONA

* Paytm Scale: 1.57 Cr device merchants, Rs 7.1L Cr GMV (Q1 FY27, Jul 2026); ~7% loan pen. (Q4 FY26, May 2026).
* Synthetic Persona: Ramesh Rao, Sri Krishna Darshini (917 scans/30d, Rs 1.75L GMV; 48 regulars silent 16-38d).

Track 1 Coverage: Growth, Operations, and Customer Recovery

PILLAR 1 : GROW BUSINESS

Retention & Dead-Hour Copilot

* **Churn Detection (Built):** SQL flags regulars silent 16+ days (16-38d in synthetic DB).
* **Dead-Hour Cohort (Built):** Identifies 3-5 PM snack buyers (55 detected, 34 eligible).
* **QR Cashback Pass (Planned):** Production hook into Paytm QR coupon redemption rail.
* **Holdout Split (Built):** Randomized 80/20 split for honest pilot measurement.

PILLAR 2 : MANAGE OPERATIONS

Rush Prep & Cashflow Signals

* **Rush Alert (Simulated):** Flags weekend morning demand spikes from scan history.
* **Supplier PO (Simulated):** Drafts pre-filled WhatsApp restock order (+18L milk, +12kg batter).
* **Cap Ledger (Stage 1):** Tracks committed cashback passes separately from API credits.
* **Credit Signal (Planned):** Feeds verified retention metrics to partner lending funnel.

PILLAR 3 : SERVE CUSTOMERS

Privacy-Safe Service Recovery

* **Delay Tagging (Simulated):** Flags peak-hour queue delays and low feedback tags.
* **Goodwill Pass (Simulated):** 1-tap bilingual apology + Rs 30 next-scan coffee voucher.
* **Zero PII Leakage (Built):** Merchant sees cohort counts/first-name hints, never raw phone numbers.
* **DPDP & Cooldown (Built):** Hard opt-in filter and 14-day per-payer message cooldown.

Closing-Time Workflow (Synthetic Dataset & Deterministic Simulation)

STEP 1 : SQL SCAN (BUILT)

Queries 1,613 synthetic UPI scans across 240 payers. Detects 48 evening regulars silent 16 to 38 days (median 19d).

STEP 2 : GUARDRAILS (BUILT)

Excludes 5 unconsented (DPDP) & 3 on 14d cooldown -> 40 eligible. Checks Rs 25/Rs 150 (16.7% <= 22% floor) & Rs 300 cap.

STEP 3 : APPROVAL (BUILT)

Plays draft Kannada/Hindi/English prompt via browser TTS; merchant approves by 1-tap button or voice command (32 nudged, 8 control).

STEP 4 : 48H SIMULATOR

Deterministic simulation: 12/32 vs 1/8 return (p = 0.24). Generates Rs 1,740 sales (5.8x sales/promo) or +Rs 222 pre-fee profit at 30% margin.

TRANSPARENCY DISCLOSURE: SYNTHETIC DATA & COMMITMENT ACCOUNTING

1. Dataset: All 240 customers and 1,613 historical UPI transactions are synthetically seeded (random.Random(20261025)); 48h redemptions are simulated.
2. Cap Commitment Note: Issuing 32 passes of Rs 25 equals Rs 800 face exposure if capped only at scan time; strict commitment accounting caps active unredeemed passes at 12 per wave for a Rs 300 hard ceiling.

Component Status: Built in MVP vs Pending vs Production Plan

TOUCHPOINTS	[BUILT] Companion PWA & Soundbox UI	[BUILT] 1-Tap Button Fallback	[PLANNED] Soundbox 4G Firmware & WhatsApp Cloud
VOICE & LLM	[BUILT] Browser Speech APIs (Draft Copy)	[PENDING] Sarvam Bulbul/Saarika Adapter	[OPTIONAL] Groq Free-Tier English Grounded Q&A
CORE ENGINE	[BUILT] Python FastAPI + SQL RFM Engine	[BUILT] 5-Rule Deterministic Policy Engine	[BUILT] 80/20 Holdout & Profit ROI Calculator
DATA & OPS	[BUILT] SQLite Ledger & Immutable Audit Log	[PLANNED] Redis Atomic Redemption Counter	[PLANNED] PostgreSQL & n8n Cron Workflows

Differentiation vs Manual Khata Apps and Unconstrained Chatbots

Dimension	Manual BI / Khata Apps	Generic LLM Chatbots	PaytmAI VyaparVoice
Merchant Interaction	Charts & manual ledger entry	Requires typing business prompts	**Proactive 8:30 PM voice brief + 1-tap button**
Customer Graph	Incomplete manual phonebooks	No link to payment history	**Paytm QR repeat-payer IDs (synthetic in MVP)**
Financial Safety	Manual coupon rules	LLM can hallucinate unsafe offers	**Deterministic 5-rule cap, floor & DPDP filter**
Privacy (DPDP)	Raw phone lists exposed	Prompt data leakage risk	**Opt-in filter; zero raw PII sent to LLMs**
ROI Measurement	Vanity message counts	No control group	**80/20 holdout + profit ROI (simulated; pilot-ready)**

Hypothesized Pilot Targets vs Simulated MVP Metrics

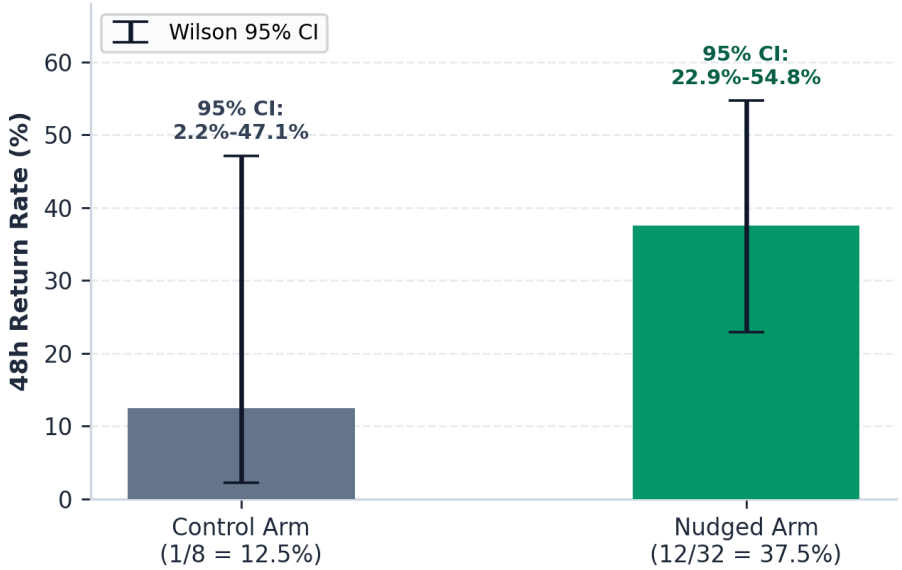
MERCHANT METRICS (SIMULATED MVP & PILOT TARGETS)

* **Simulated Profit ROI:** +Rs 222 pre-fee (+74%) / +Rs 91.50 post-fee at 30% margin (5.8x sales/promo).
* **Repeat Visit Target:** +14% to +22% monthly repeat visits (PILOT TARGET - unvalidated).
* **Waste Reduction Target:** -18% afternoon perishable loss (PILOT TARGET - unvalidated).

PAYTM ECOSYSTEM HYPOTHESES (PILOT TARGETS)

* **Device Retention:** -35% Soundbox churn reduction (PILOT TARGET - unvalidated).
* **Store QR Share:** +12% to +15% store Paytm QR GMV lift (PILOT TARGET - unvalidated).
* **Credit Funnel:** Expand ~7% loan pen. (Q4 FY26) toward 15% (LONG-TERM TARGET).

SIMULATED Holdout + 95% Wilson CIs (p = 0.24)



1.74x

Gross Profit / Promo
(30% Margin Sim.)

p = 0.24

Fisher Exact (N=40)
Needs ~46 / Arm

Rs 5.00

Cached AI Cost/Mo
(Rs 28.80 Uncached)

Sensitivity Across Revenue Streams and TTS Caching Rates

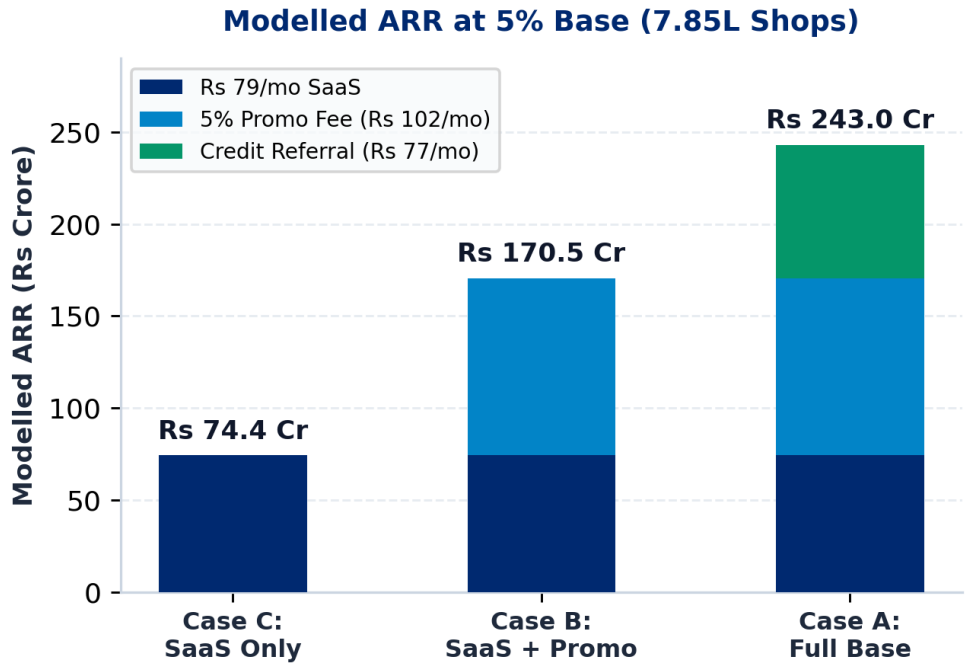
* **Stream 1 (SaaS - Rs 79/mo):** Bundled on Soundbox billing (~Rs 90-100/mo, FY25 AR) after 2 free trial campaigns -> Rs 74.4 Cr ARR at 5% base.

* **Stream 2 (Promo Fee - Rs 102/mo):** 5% fee on ~Rs 2,040/mo redeemed promo bills -> +Rs 96.1 Cr (Rs 170.5 Cr ARR combined with SaaS).

* **Stream 3 (Credit Uplift - Rs 77/mo):** 1.5%-2.5% partner NBFC origination fee uplift -> +Rs 72.5 Cr (Rs 243.0 Cr Full Base ARR).

* **Cached COGS (Rs 48.00/mo):** Rs 1.80 SLM + Rs 3.20 TTS (~88% cache hit on Bulbul v3) + Rs 43 messaging = 81.4% gross margin on Rs 258 ARPU.

* **Uncached Sensitivity (Rs 71.80/mo):** Uncached Bulbul v3 (9k chars/mo at Rs 30/10k = Rs 27.00 TTS) = 72.2% gross margin (Bulbul v2 Rs 13.50 = 77.4%).



CONSERVATIVE PRESENTATION ANCHOR

Lead with Rs 74.4 Cr (SaaS-only) to Rs 170.5 Cr (Core Product) ARR at 5% adoption; treat Rs 243.0 Cr (with credit referrals) as upside.

APPENDIX A9 : VALIDATION & ROLLOUT ROADMAP

From Synthetic Hackathon Slice to Powered 50-Store Bengaluru Pilot

STAGE A : BUILT TODAY (MVP)

Synthetic Slice & Guardrails

* FastAPI + SQLite synthetic store (240 payers, 1,613 scans)

* 5-rule policy engine + 80/20 holdout & profit ROI math

* Companion Soundbox UI + 21/21 automated QA checks

STAGE B : DAYS 1 TO 45 PILOT

Bengaluru 50-Store Validation

* Verify Sarvam TTS/ASR in sponsor window & native copy review

* Run >=4 weekly waves (~50+ payers/arm) for 80% statistical power

* Measure real net merchant profit after cashback & 5% fee

STAGE C : PRODUCTION SCALE

Paytm Ecosystem Integration

* Integrate Paytm QR dynamic coupon ledger & 4G Soundbox push

* Self-host open-weight SLMs (e.g., Apache 2.0 Sarvam-30B/M) if validated

* Roll out across Soundbox languages & partner credit funnel